Sensor requirements and reliability

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Temperature could be omitted under a predefined benchmark rule when heart rate was available. A low overall error still concealed a long-session reliability gap.

0.978	6 vs 7	79.54%
primary test MAE*	required input fields	long-session interval coverage

A controlled comparison

The supplied 15,000-record CSV yielded 14,999 observations after one exact duplicate was removed. IDs were excluded from inputs. Fixed partitions contain 8,999 development, 3,000 calibration and 3,000 test records; five development folds are shared across comparisons.

A mean baseline, four Ridge pipelines and four XGBoost pipelines were compared. Each XGBoost input set received the same 30-configuration search. Calibration separates 1,500 interval-fit from 1,500 interval-audit records. Model selection and thresholds were frozen before a one-time final test release; test results did not retune the system.

Frozen XGBoost results

Available inputs	Fields	Test MAE*	Test RMSE*
Duration only	1	11.508	16.953
Basic measurements	5	8.240	11.723
Basic + heart rate (primary)	6	0.978	1.576
Full, including temperature	7	1.032	1.645

*Against the same 3,000 recorded labels, in interpreted kcal/session. Original units and label construction are unverified.

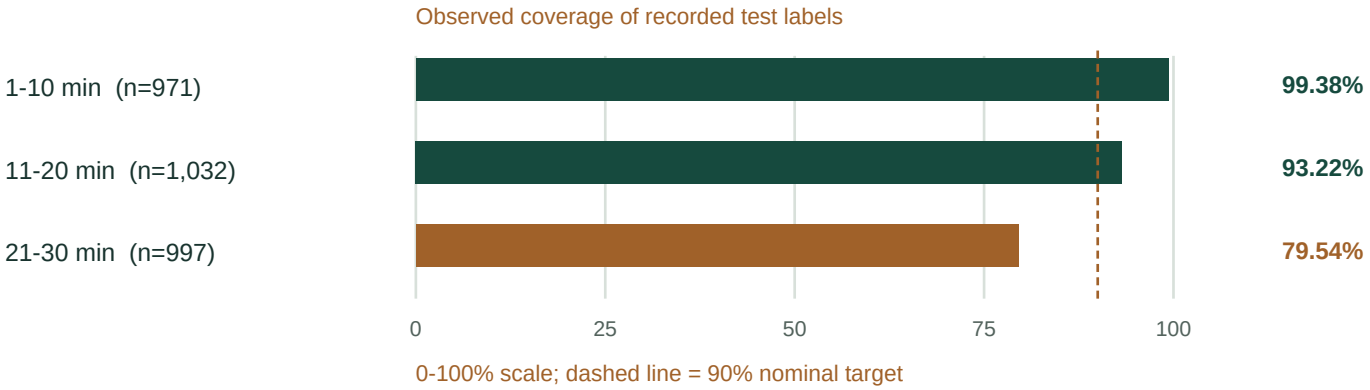
The heart-rate model's relative MAE change versus full inputs is **-5.22%**; approximate paired 95% percentile interval **[-6.52%, -3.82%]** using 2,000 row resamples. Its upper bound passes the project's +10% tolerance. Removing both sensors fails. One fewer field is an input-count result, not measured cost or time savings.

Scope: previously explored benchmark; unknown collection, possible participant dependence and unresolved dataset licensing. This is not verified physiological calorie measurement.

RELIABILITY / IMPLEMENTATION / EVIDENCE

Make the failure cases visible

The primary fixed 90%-nominal interval covers 90.67% overall. Coverage differs substantially by session duration; an overall score does not establish reliability for every group.



For 21-30 minute sessions (997 records), coverage is only **79.54%**. Bounds remain unclipped; no subgroup recalibration followed the test. An earlier simulated Gaussian heart-rate-noise stress test (SD 5 bpm) raised development MAE from 1.080 to 7.777. That is a chosen simulation, not a measured device error distribution.

A deployed, verifiable implementation

The free Hugging Face Static Space runs four exported frozen forests in the browser. Feature order, binary one-hot encoding, float32 comparisons and summation reproduce the Python pipelines. The app sends no prediction inputs to a prediction API and saves no submissions; hosting may retain asset-request logs.

Served files and repository hashes match the reviewed package. Downloaded JavaScript and models match Python predictions and unclipped bounds exactly on **5,854 synthetic sessions**, including split boundaries. The current source passes **82 Python tests and four JavaScript checks**. The author confirmed basic demo interaction; automated browser/mobile/accessibility checks were unavailable.

This demonstrates browser model delivery and deployment verification. A Python/Waitress Docker template is retained locally, but Linux container execution and hosted Docker deployment were not verified.

Authorship and what is not measured

The author supplied the existing notebook/data, directed project choices, selected free hosting, completed account authentication and checked basic live interaction. Research code, evaluation tooling and documentation were produced with substantial AI-assistant support. Project ownership does not establish independent mastery of every implementation detail.

The usability study was skipped at the author's request: **no participant results or measured user benefit** exist. No clinical validity, adoption, health improvement, cost savings or new algorithm is claimed. Row bootstrap intervals assume exchangeability and omit training uncertainty. A documented measured-data replication would be a separate future study.

Explore the evidence

Live demo: huggingface.co/spaces/Moobakr/calorie-lab-fewer-sensors
Project source: github.com/Moobakr/calorie-lab-research

The repository and accompanying source archive contain the evaluation protocol, final and reliability reports, aggregate metrics, deployment checks and an explicit contribution record.